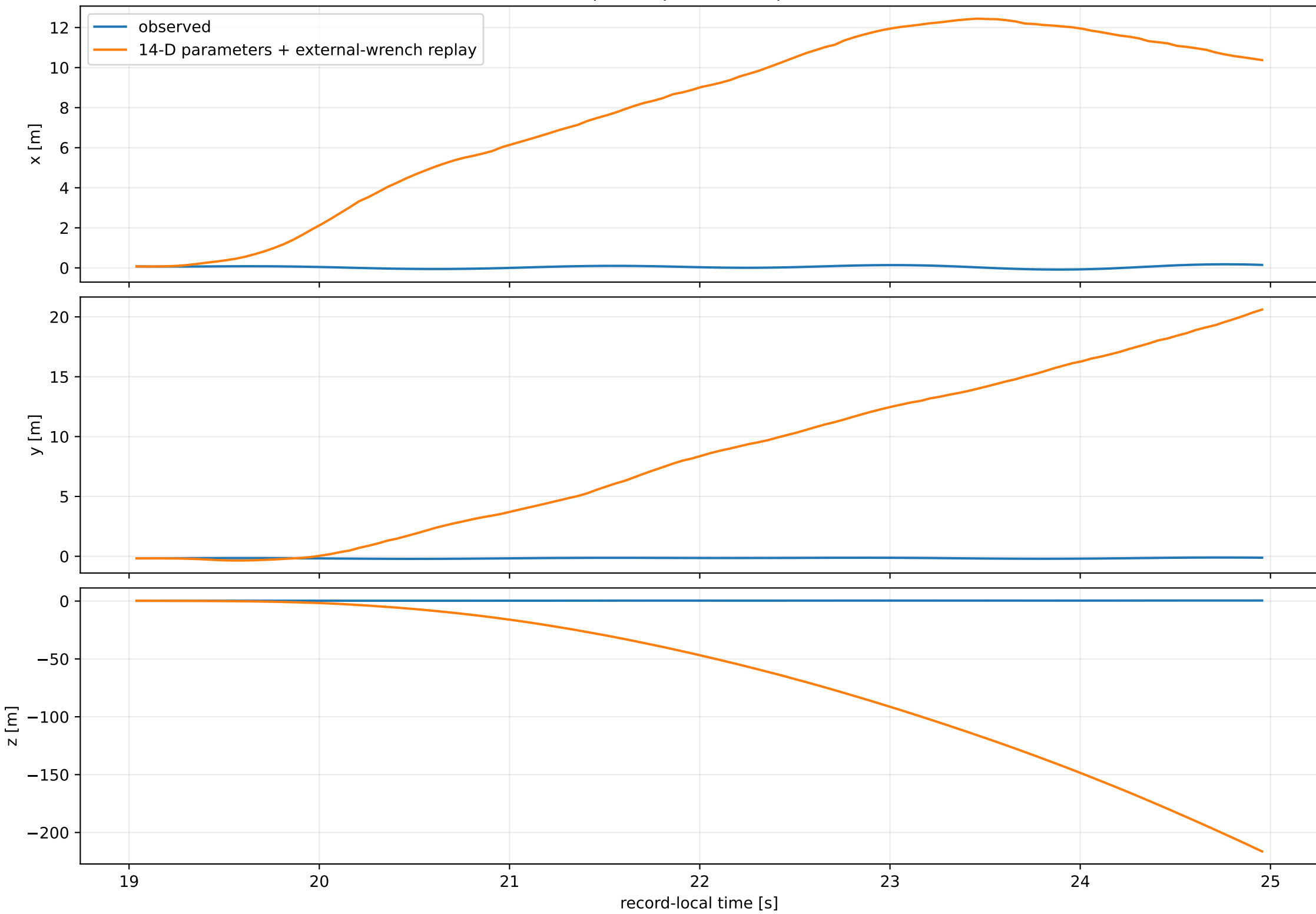
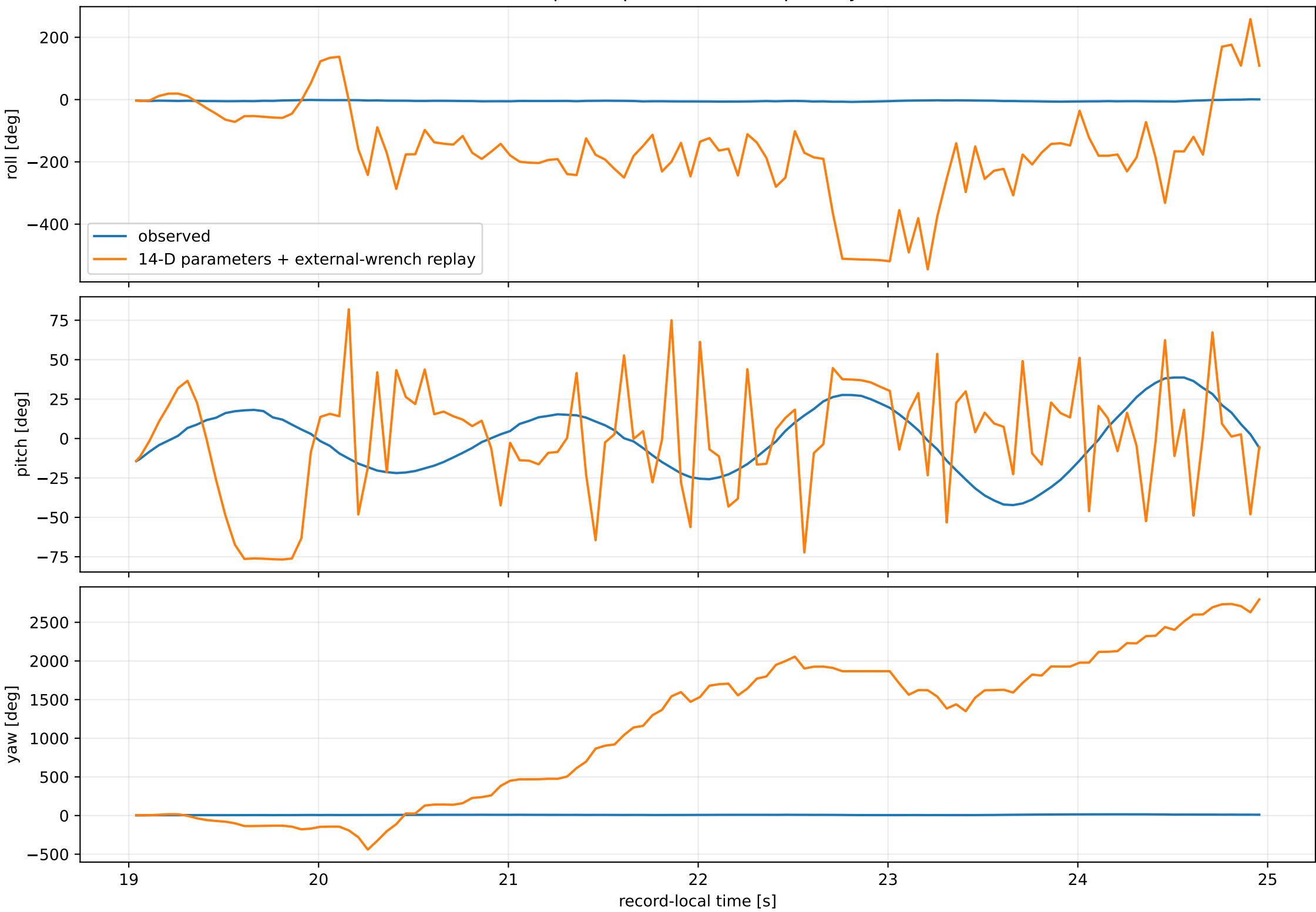


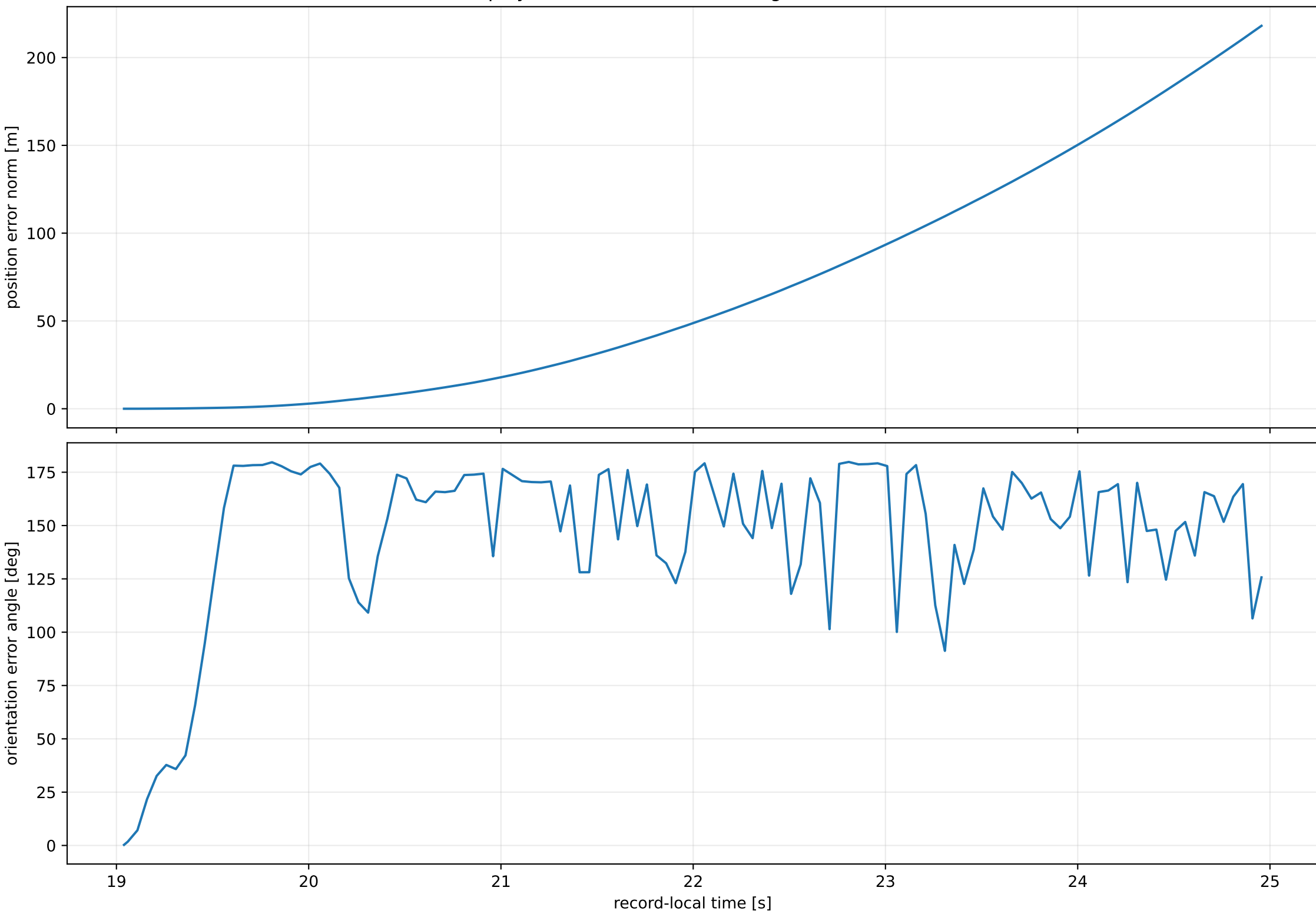
Fine pose reproduction: position



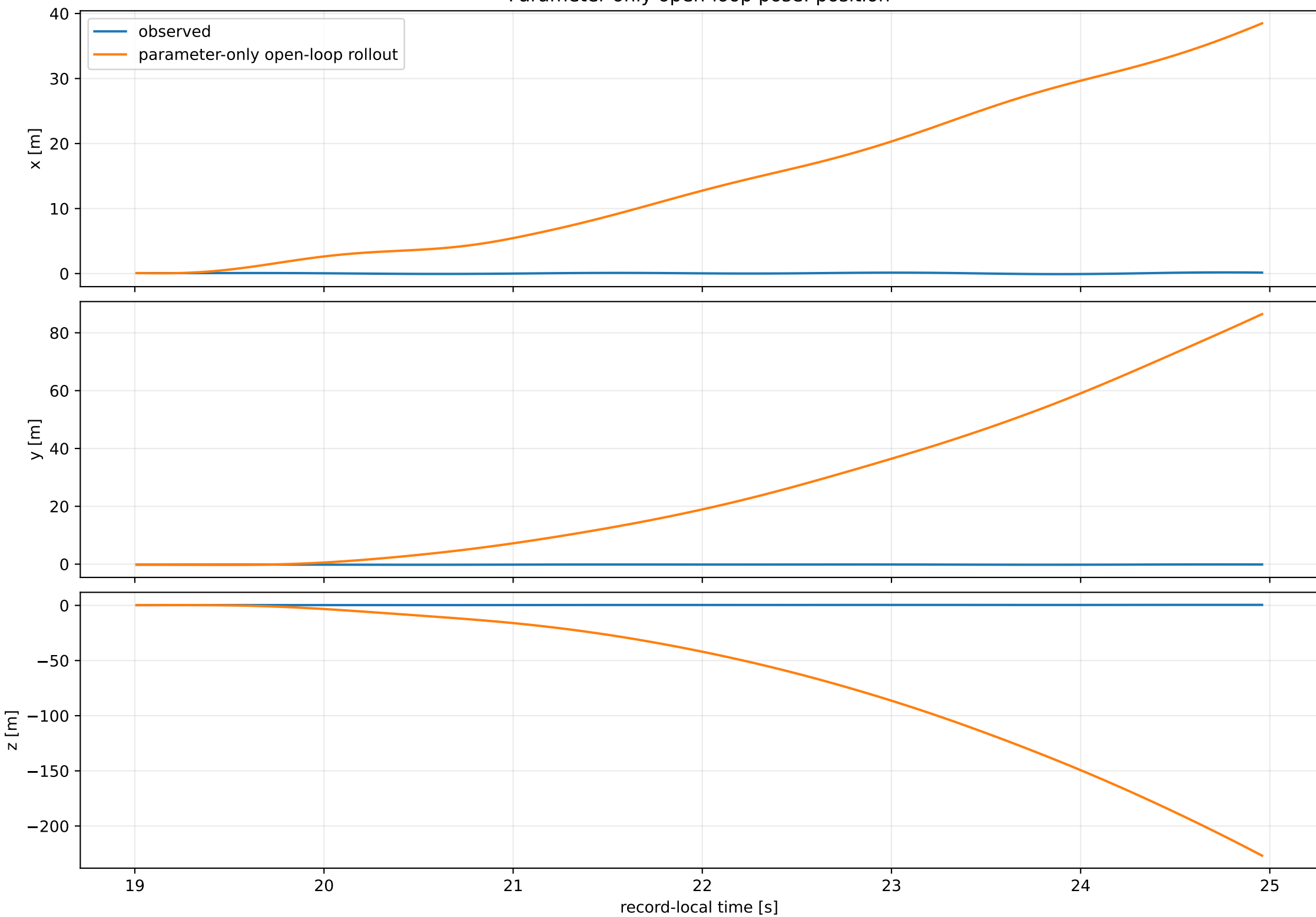
Fine pose reproduction: roll / pitch / yaw



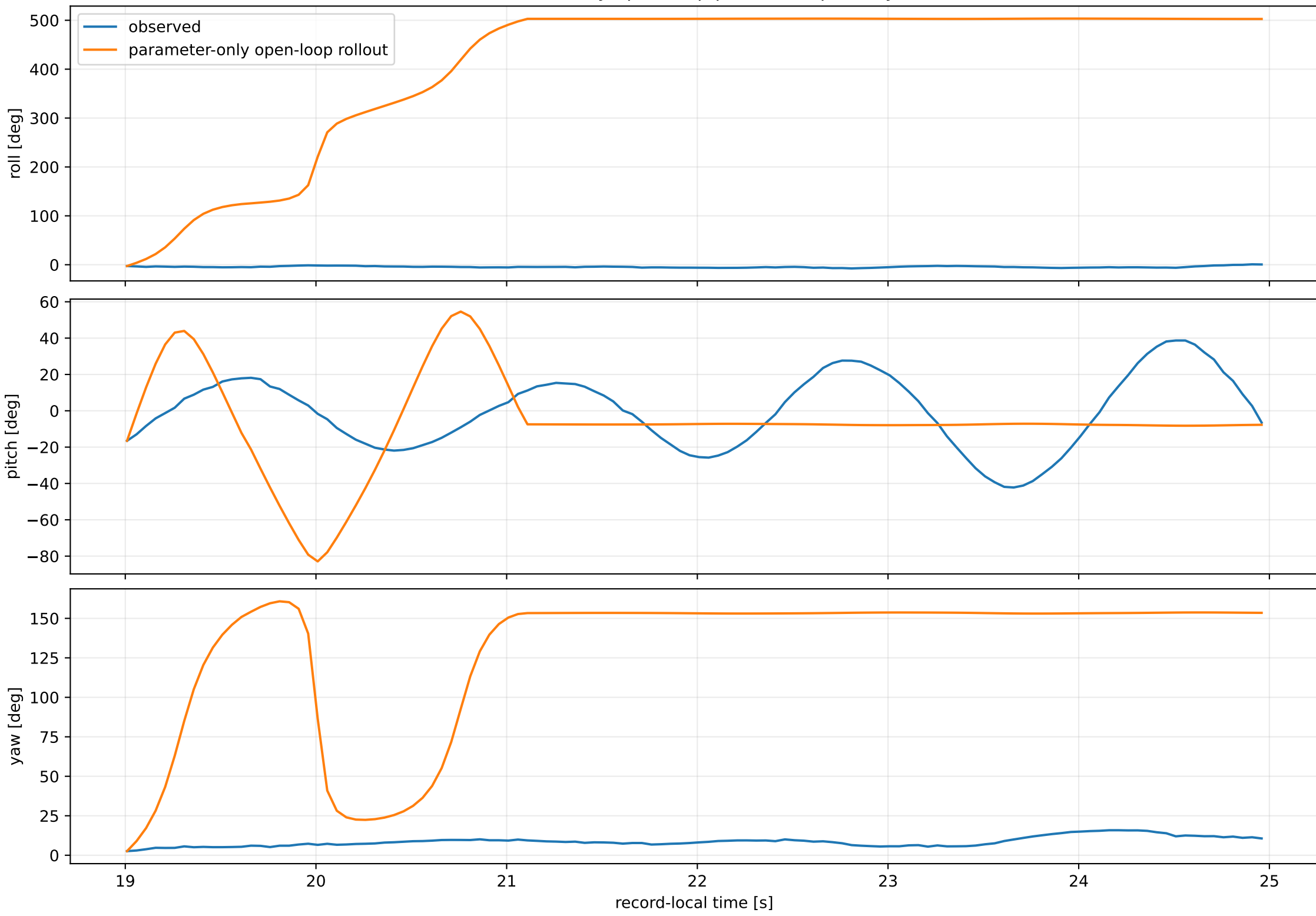
Fine replay error after the 14-D change — lower is better



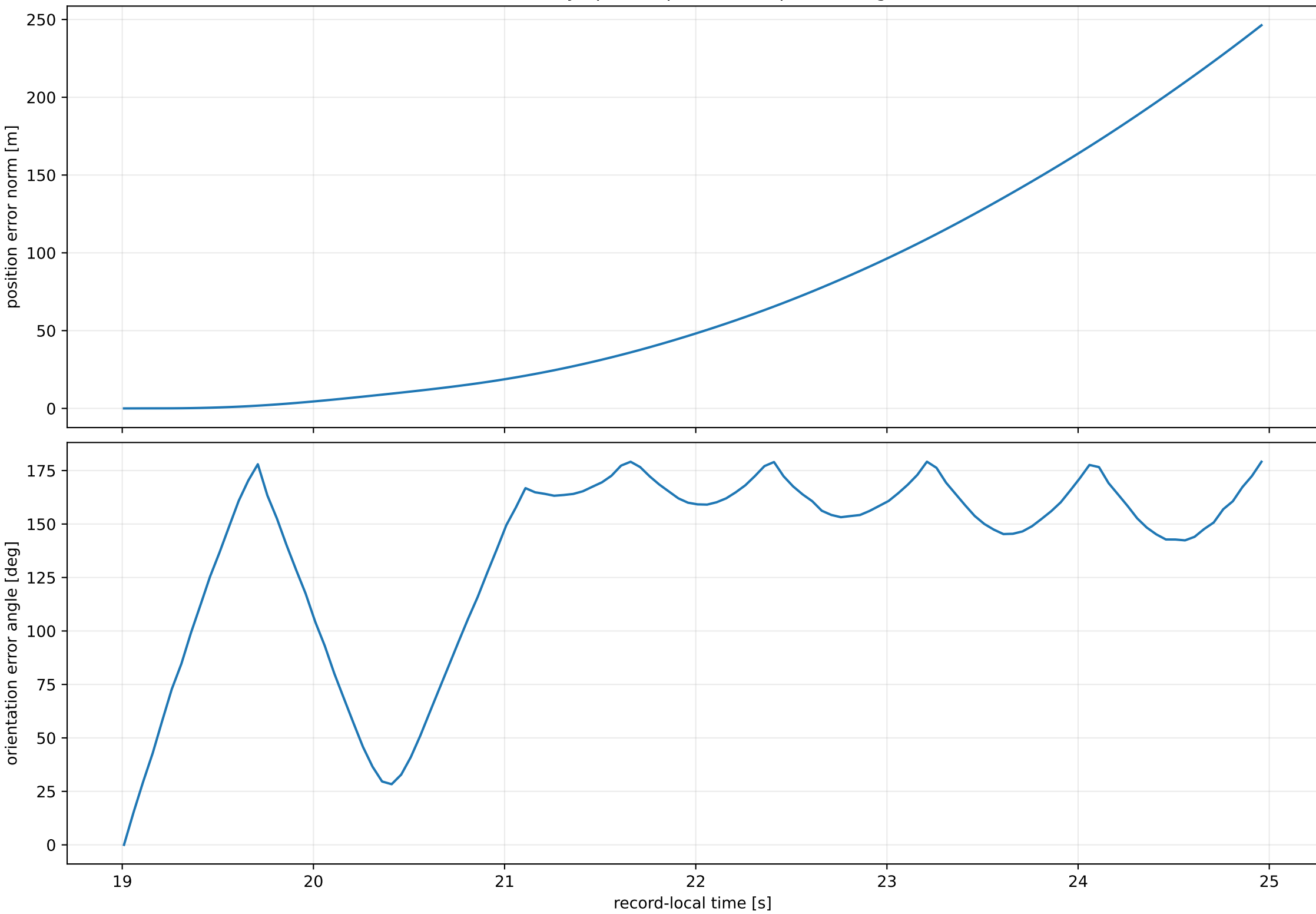
Parameter-only open-loop pose: position



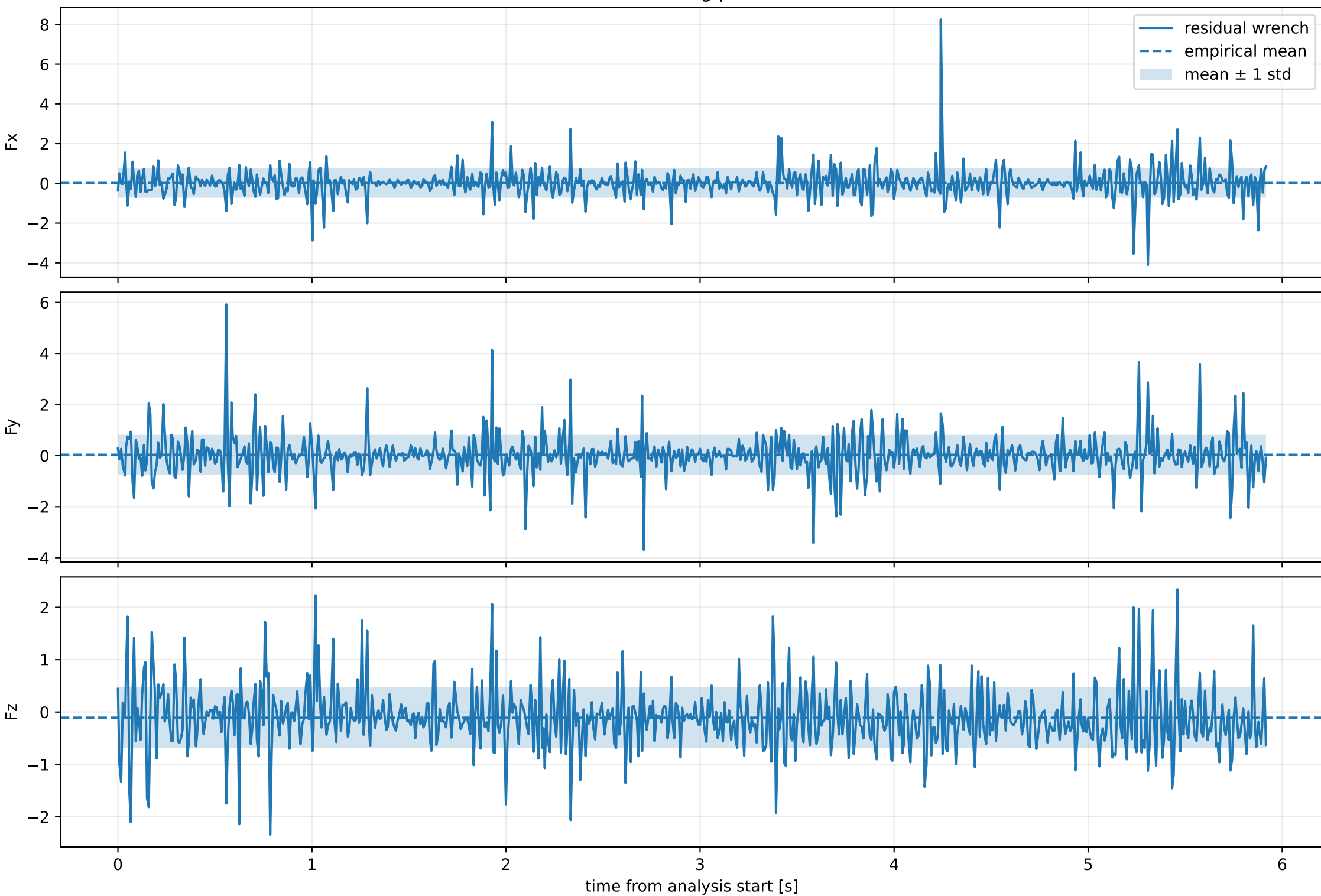
Parameter-only open-loop pose: roll / pitch / yaw



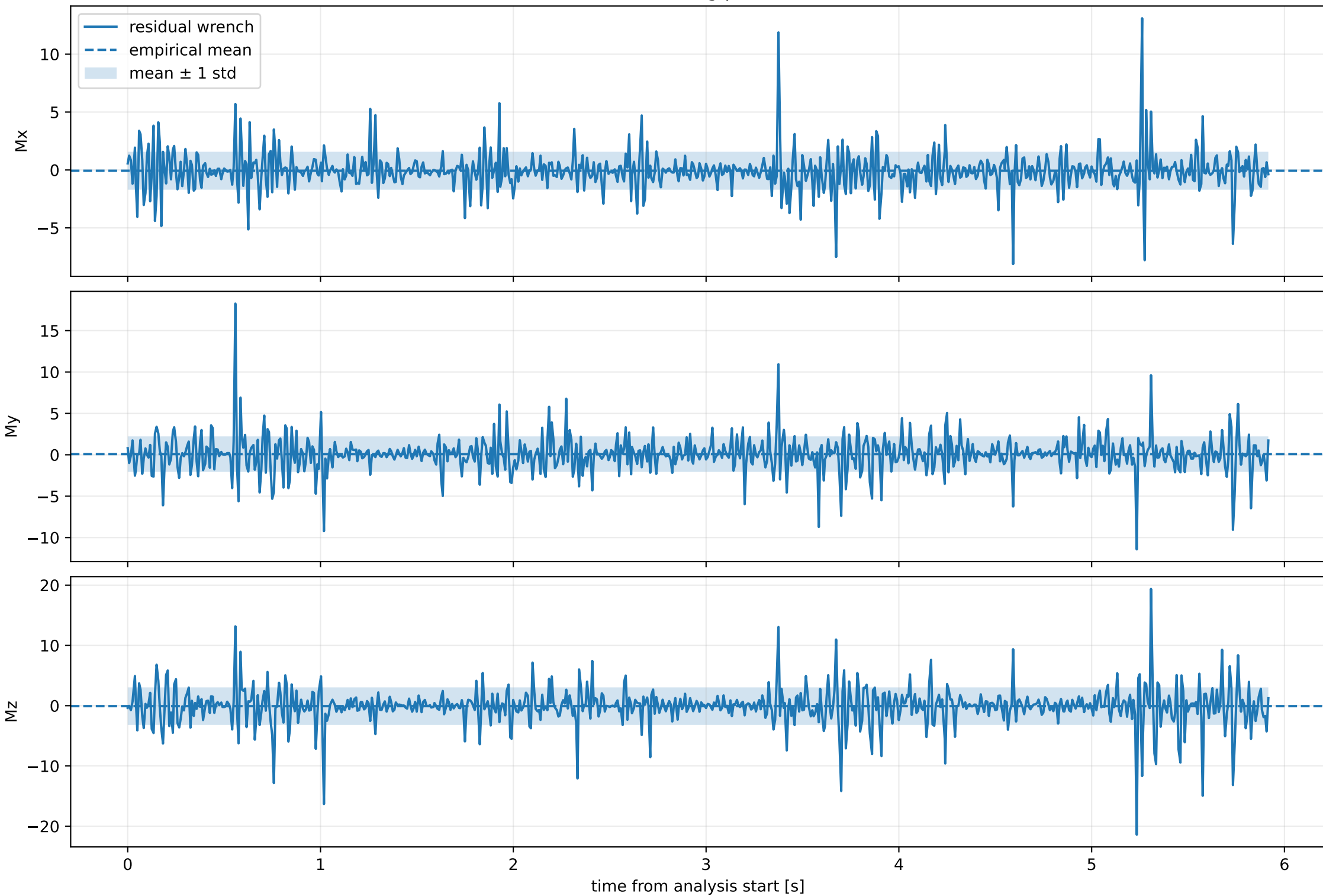
Parameter-only open-loop rollout (separate diagnostic)



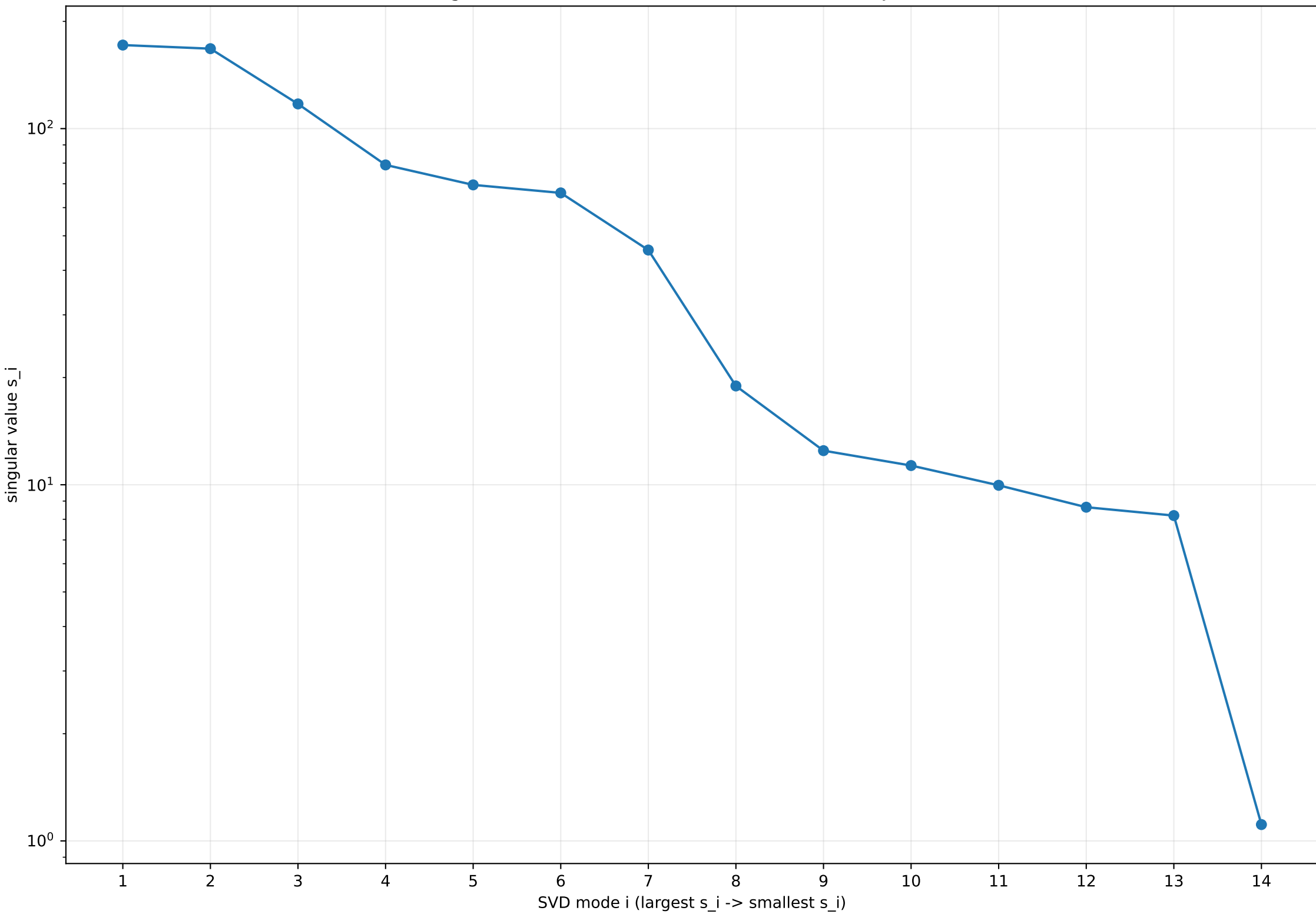
Dimensionless residual force: empirical mean  $\pm$  1 std  
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



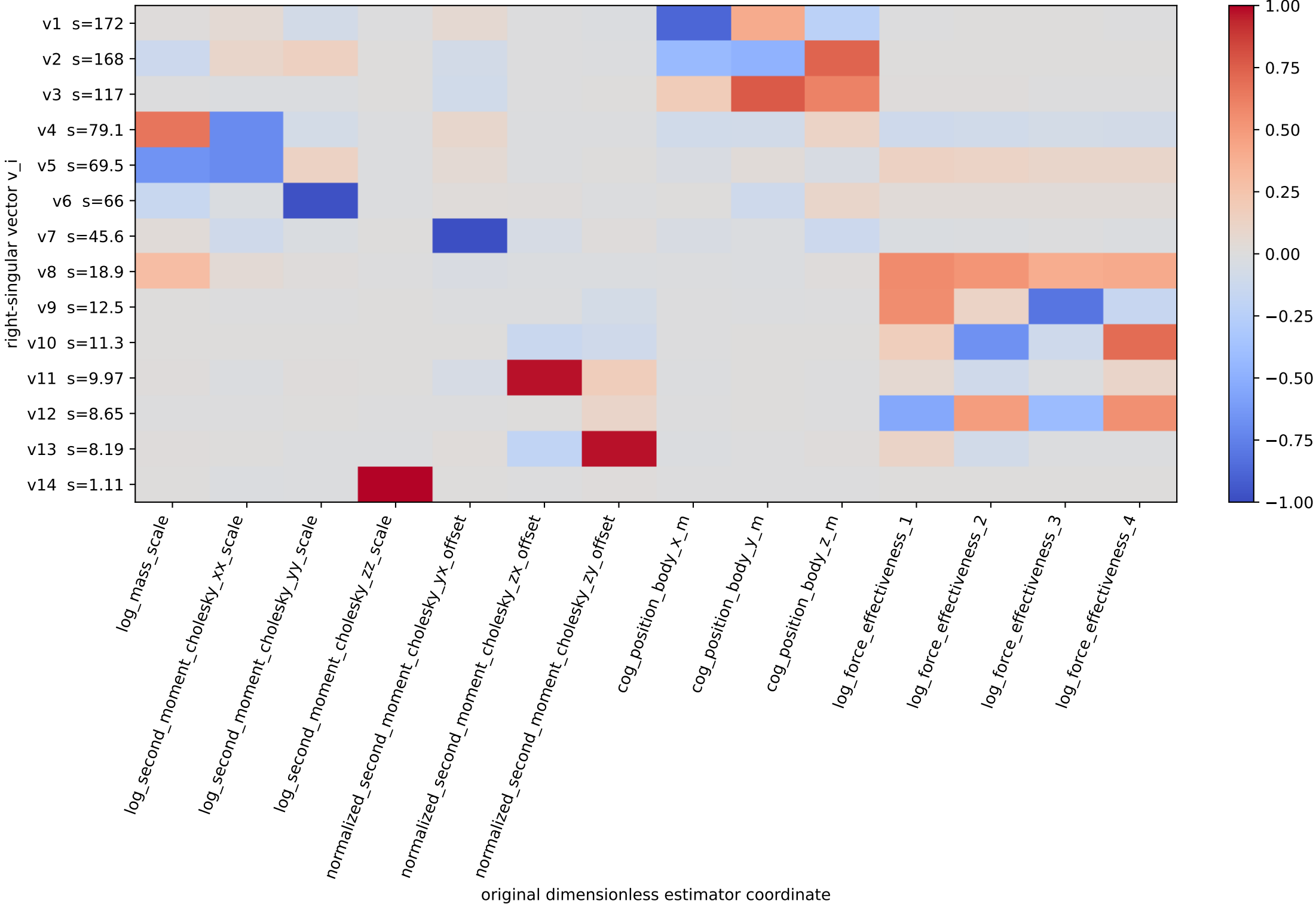
Dimensionless residual torque: empirical mean  $\pm$  1 std  
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors  $V^T$   
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36464523465

inertia J [kg m<sup>2</sup>]

[ 0.065121594 -0.00079752156 -0.00014751371]  
[-0.00079752156 0.065950919 -0.00065553356]  
[-0.00014751371 -0.00065553356 0.12877604]

principal moments [kg m<sup>2</sup>]

[ 0.06463398, 0.06643139, 0.12878318]

CoG position in body frame [m]

[-0.02352233, -0.027895114, 0.047646334]

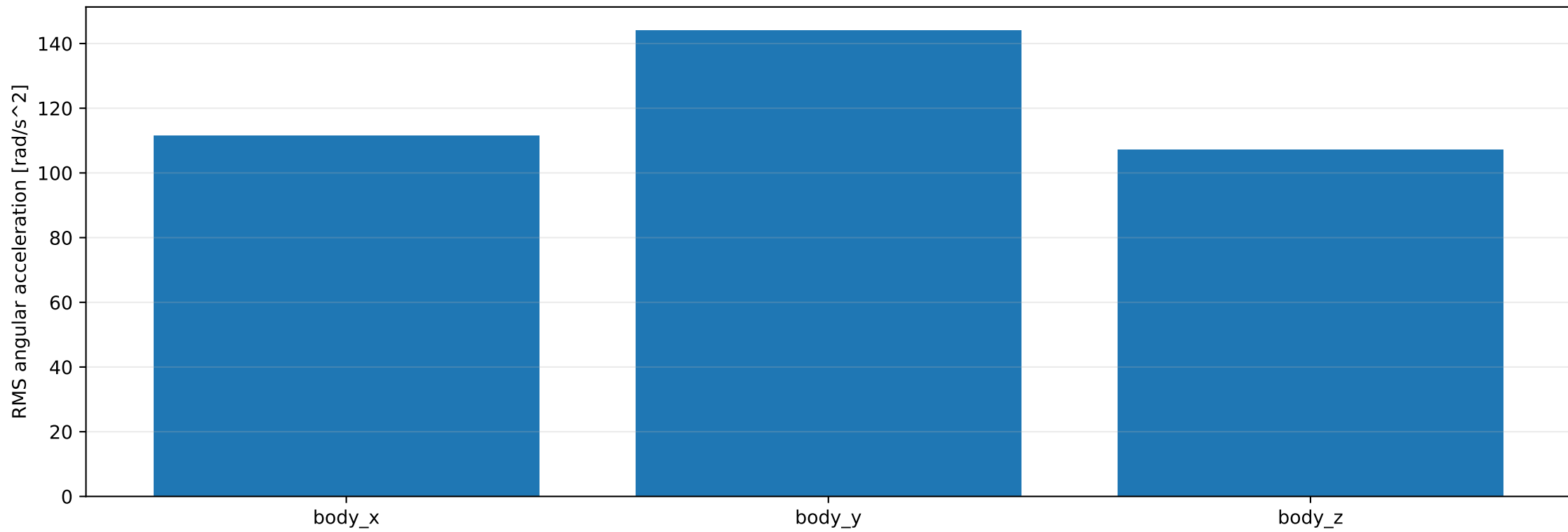
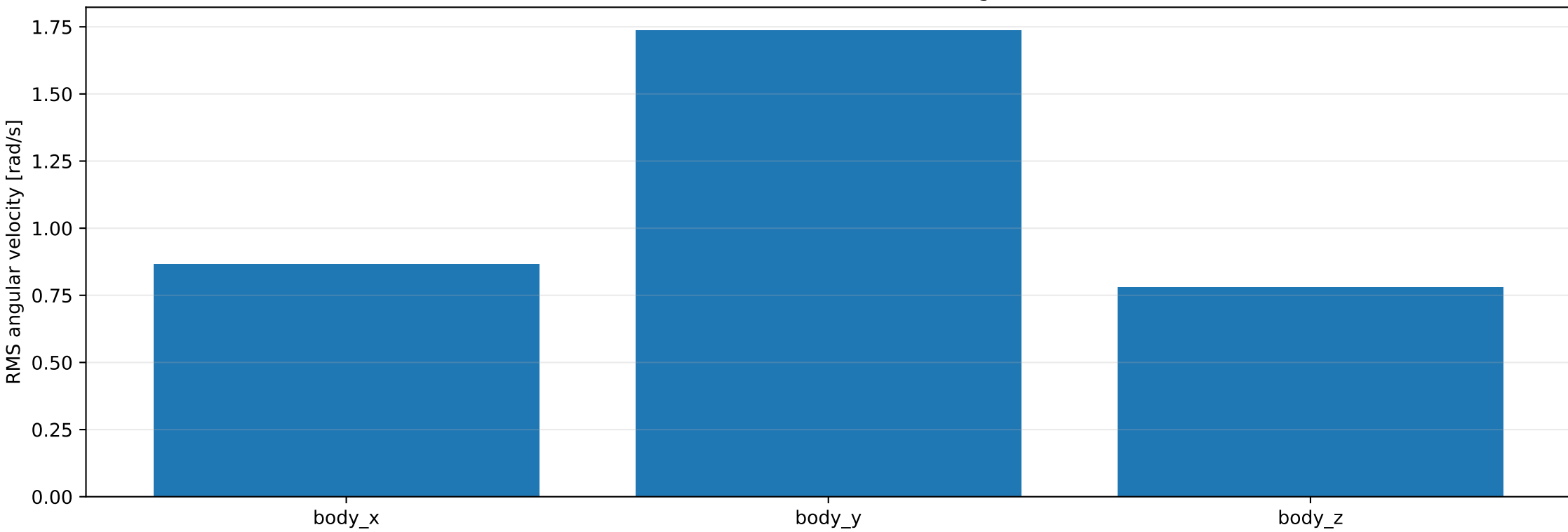
rotor force effectiveness

[ 1.0353556, 0.96444846, 0.88936589, 0.95150605]

selected delay [s]

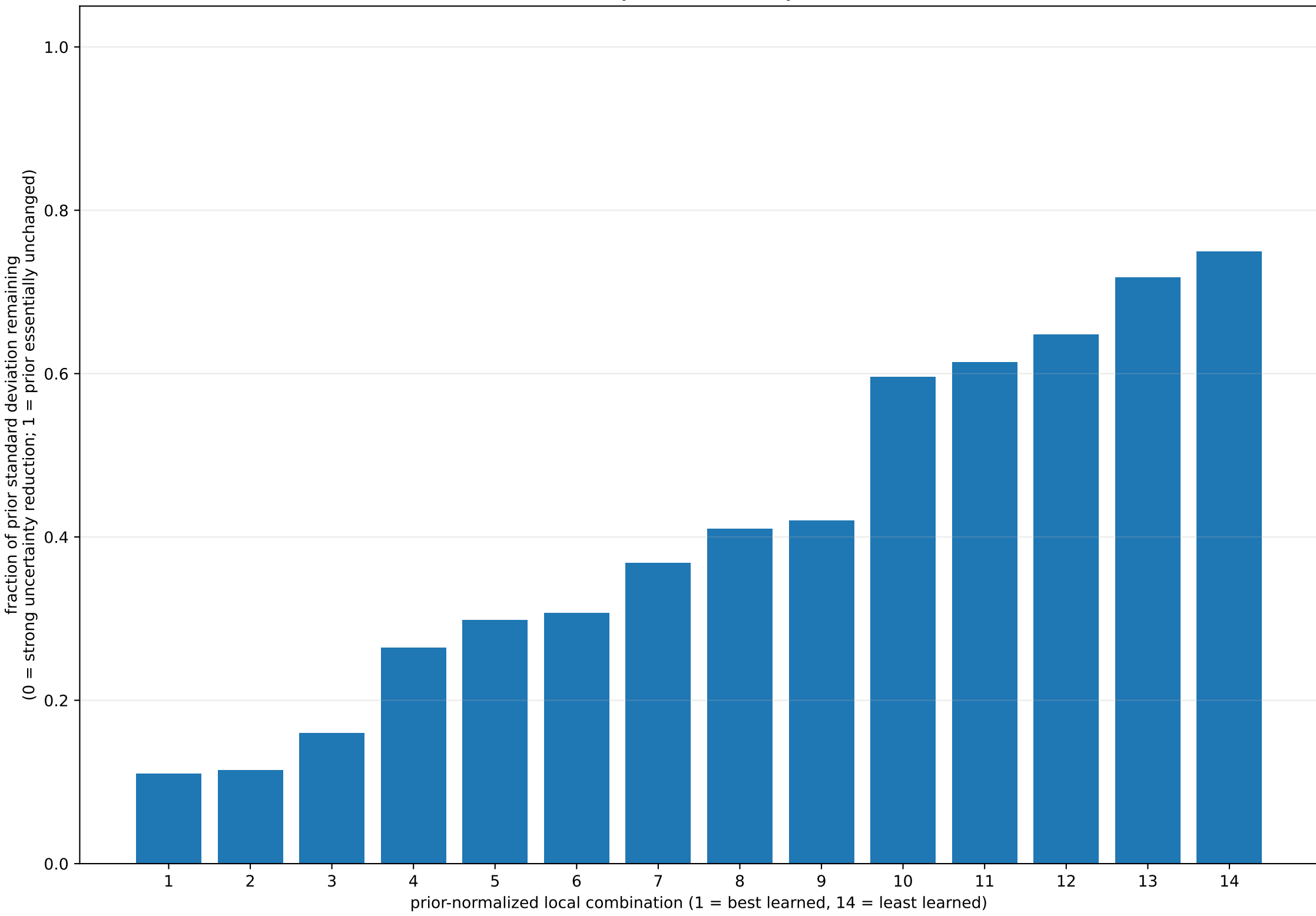
0.190004539397

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

